SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	SWTID-2026-7714
Project Title	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to build an intelligent credit card approval prediction system using machine learning to improve decision accuracy, reduce manual effort, and enhance customer experience. The system will analyze applicant data and provide fast, transparent, and reliable approval predictions.

Project Overview	
Objective	The primary objective is to develop a machine learning based system that predicts the likelihood of credit card approval for applicants based on their personal, financial, and employment information.
Scope	The project includes data collection, data preprocessing, exploratory data analysis, feature engineering, model building, evaluation, and deployment of a web application for real-time credit card approval prediction.
Problem Statement	
Description	Traditional rule-based methods for credit card approval are time-consuming, subjective, and may lead to inaccurate or inconsistent decisions. There is a need for a data-driven solution that can evaluate multiple factors and predict approval more accurately.
Impact	The proposed system will improve approval accuracy, reduce rejection of eligible applicants, minimize default risk, save time and operational costs, and enhance customer satisfaction through faster and transparent decisions.
Proposed Solution	
Approach	Implement a machine learning model using historical applicant data to predict credit card approval. The system will provide prediction results along with key factors influencing the decision and a user-friendly interface for applicants and administrators.
Key Features	• Accurate credit card approval prediction using ML models • Considers multiple applicant and financial factors • Provides reason/explanation for approval or rejection • User-friendly web interface for applicants • Admin dashboard for monitoring applications and model performance • Reduces manual verification time and improves efficiency